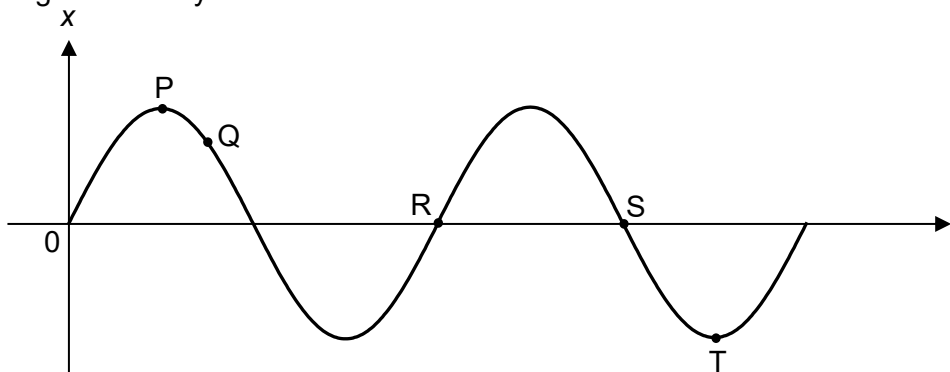


16 The graph below shows the variation with time t of the displacement x of a simple harmonic oscillator oscillating horizontally.



Which statement is correct?

- A** The phase difference between P and S is $\frac{\pi}{2}$ rad.
- B** The acceleration at T is zero.
- C** The potential energy of the oscillator is larger at Q compared to T.
- D** The kinetic energy of the oscillator is larger at Q compared to R.

Ans: A

A-phase difference is $\frac{\Delta t}{T} \times 2\pi = \frac{\left(\frac{T}{4}\right)}{T} \times 2\pi = \frac{\pi}{2}$

B- at T acceleration is maximum (max. displacement)

C-maximum PE at T ($x_{\max}$), lesser PE at Q

D-maximum KE at R (equilibrium position), lesser KE at Q

17 The diagram below shows three polarisers W , X and Y